

Kamogelo Makhura

Week 5 Practical

Tasks:

Part A: OSPF Multi-Area

1. Address the R-EDGE-R3 link and R3's LAN.
2. On R-EDGE (now the ABR): add network [R-EDGE-R3 link] [wildcard] area 1

The image displays two screenshots from Cisco Packet Tracer. The top screenshot shows the 'R-EDGE' router configuration window. The bottom screenshot shows the 'Router3' configuration window.

R-EDGE Configuration:

```
Router>enable
Router#config t
Router(config)#interface g0/0
Router(config-if)#ip address 10.0.23.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
%LINK-6-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#
Router(config)#interface g0/1
Router(config-if)#ip address 10.0.23.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
%LINK-6-CHANGED: Interface GigabitEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#
```

Router3 Configuration:

```
Router3>enable
Router3#config t
Router3(config)#interface g0/0
Router3(config-if)#ip address 10.0.23.2 255.255.255.252
Router3(config-if)#no shutdown
Router3(config-if)#
%LINK-6-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router3(config-if)#exit
Router3(config)#
Router3(config)#interface g0/1
Router3(config-if)#description R3 lan
Router3(config-if)#ip address 192.168.100.1 255.255.255.0
Router3(config-if)#no shutdown
Router3(config-if)#
%LINK-6-CHANGED: Interface GigabitEthernet0/1, changed state to up
Router3(config-if)#exit
Router3(config)#
```

Cisco Packet Tracer - D:\Year 2\Semester 1\Practicals\WEEK 3.plt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 605, y: 178

Time: 00:18:22

Search for device

Automatically Choose Connection Type

R-EDGE

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#ip address 10.0.23.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
%LINK-3-CHANGED: Interface GigabitEthernet0/1, changed state to up
Router(config)#
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router(config)#end
Router#
$SYS-S-CONFIG_I: Configured from console by console
Router#
Router#
Router#
Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 10.0.23.0 0.0.0.3 area 1
Router(config-router)#exit
Router#
$SYS-S-CONFIG_I: Configured from console by console
Router#write
Building configuration...
[OK]
Router#
Router#
Router#
```

Copy Paste

Top

Router3

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#description R3 lan
Router(config-if)#ip address 192.168.100.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#end
Router#
$SYS-S-CONFIG_I: Configured from console by console
Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#router-id 3.3.3.3
Router(config-router)#network 10.0.23.0 0.0.0.3 area 1
Router(config-router)#network 192.16
00:19:38: *OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
Router(config-router)#network 192.168.100.0 0.0.0.255 area 1
Router(config-router)#exit
Router#
$SYS-S-CONFIG_I: Configured from console by console
Router#write
Building configuration...
[OK]
Router#
Router#
Router#
```

Copy Paste

Top

Root

2911 Router3

Realtime Simulation

Scenario 0

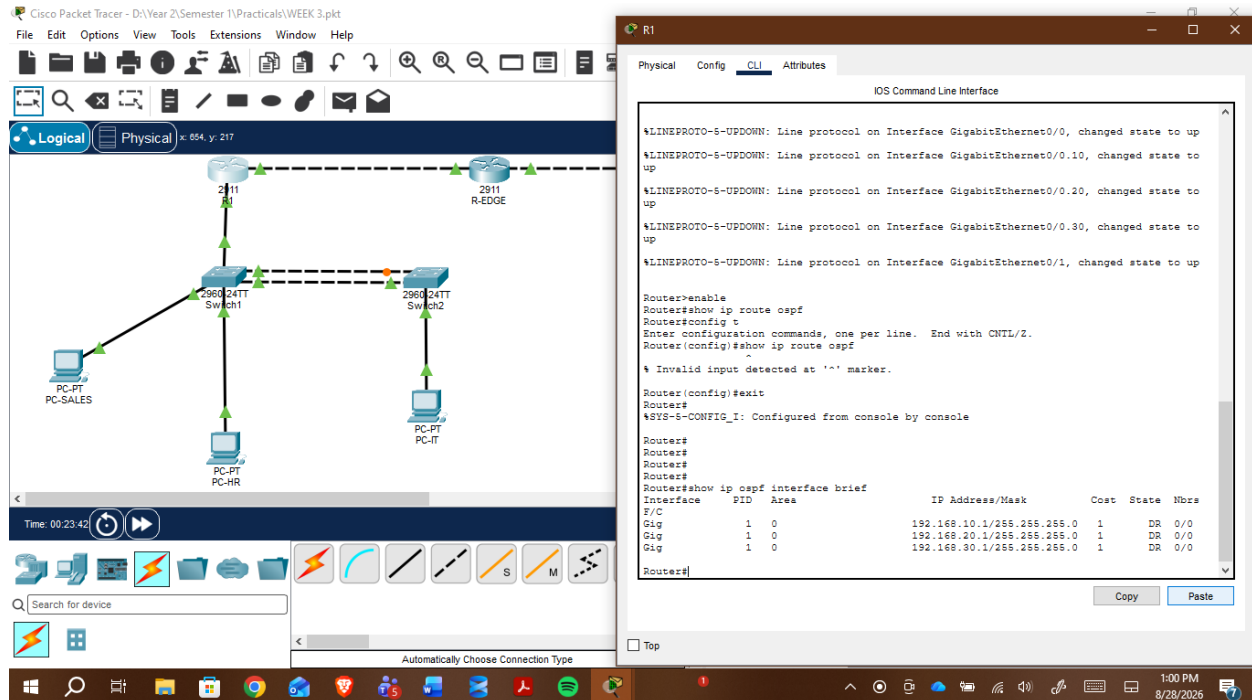
New Delete

Toggle PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
------	-------------	--------	-------------	------	-------	-----------	----------	-----

12:57 PM 8/28/2026

1. On R3: router ospf 1 network [R-EDGE-R3 link] [wildcard] area 1 network [R3's LAN] [wildcard] area 1
2. show ip route ospf — confirm inter-area routes appear marked O IA.



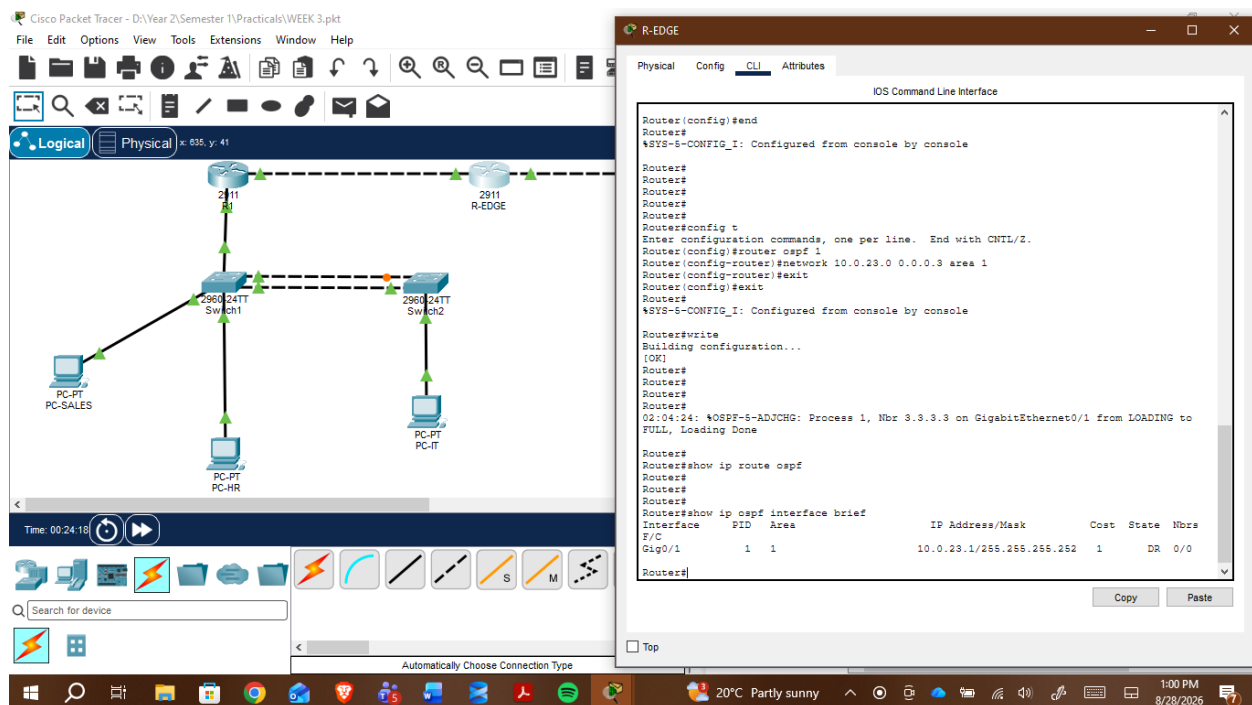
The network diagram shows a central 2911 R-EDGE router connected to two 2960 24TT switches (Switch1 and Switch2). Switch1 is connected to three PCs (PC-SALES, PC-PT, PC-HR) and Switch2 is connected to two PCs (PC-PT, PC-IT). The R-EDGE router is also connected to a 2911 R3 router. The R1 CLI window shows the following commands and output:

```

Router>enable
Router#show ip route ospf
Router#config t
Router(config)#show ip route ospf
Router(config)#
Router#
Router#
Router#
Router#show ip route ospf interface brief
Router#

```

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs
Gig	1	0	192.168.10.1/255.255.255.0	1	DR	0/0
Gig	1	0	192.168.20.1/255.255.255.0	1	DR	0/0
Gig	1	0	192.168.30.1/255.255.255.0	1	DR	0/0



The network diagram is the same as the one above. The R-EDGE CLI window shows the following commands and output:

```

Router(config)#end
Router#
Router#
Router#
Router#
Router#
Router#config t
Router(config)#router ospf 1
Router(config-router)#network 10.0.23.0 0.0.0.3 area 1
Router(config-router)#exit
Router(config)#exit
Router#
Router#
Router#
Router#show ip route ospf
Router#
Router#
Router#
Router#show ip route ospf interface brief
Router#

```

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs
Gig0/1	1	1	10.0.23.1/255.255.255.252	1	DR	0/0

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File Edit Options View Tools Extensions Window Help

Logical Physical x: 1458, y: 312

Time: 00:48:18

Search for device

Automatically Choose Connection Type

R-C

Physical Config CLI Attributes

IOS Command Line Interface

```
R-C(config-if)#  
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up  
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up  
R-C(config-if)#exit  
R-C(config)#  
R-C(config)#interface loopback0  
R-C(config-if)#  
%LINK-5-CHANGED: Interface Loopback0, changed state to down  
%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up  
R-C(config-if)# ip address 192.168.3.1 255.255.255.0  
R-C(config-if)#exit  
R-C(config)#  
R-C#  
%SYS-5-CONFIG_I: Configured from console by console  
R-C#  
R-C#  
R-C#ping neighbor IPs  
* Invalid input detected at '^' marker.  
R-C#  
R-C#conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
R-C(config)#router eigrp 100  
R-C(config-router)# network 10.0.0.0  
R-C(config-router)# network 192.168.1.0  
R-C(config-router)# network 192.168.2.0  
R-C(config-router)# network 192.168.3.0  
R-C(config-router)# no auto-summary  
R-C(config-router)#exit  
R-C(config)#  
R-C#
```

R-A

Physical Config CLI Attributes

IOS Command Line Interface

```
R-A(config)#end  
R-A#  
%SYS-5-CONFIG_I: Configured from console by console  
R-A#  
R-A#conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
R-A(config)#router eigrp 100  
R-A(config-router)# network 10.0.0.0  
R-A(config-router)# network 192.168.1.0  
R-A(config-router)# network 192.168.2.0  
R-A(config-router)# network 192.168.3.0  
R-A(config-router)# no auto-summary  
R-A(config-router)#exit  
R-A(config)#  
R-A#  
%DUAL-5-NBCHANGE: IP-EIGRP 100: Neighbor 10.1.12.2 (GigabitEthernet0/0) is up: new adjacency  
R-A(config)#  
R-A(config)#end  
R-A#  
%SYS-5-CONFIG_I: Configured from console by console  
R-A#  
R-A#show ip eigrp neighbors  
IP-EIGRP neighbors for process 100  
H Address Interface Hold Uptime SRTT RTO Q Seq  
0 10.1.12.2 Gig0/0 12 00:03:29 40 1000 0 5  
R-A#  
R-A#show ip route eigrp  
10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks  
D 10.1.23.0/30 [90/3072] via 10.1.12.2, 00:03:50, GigabitEthernet0/0  
D 192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks  
D 192.168.2.0/24 [90/130816] via 10.1.12.2, 00:03:50, GigabitEthernet0/0  
R-A#
```

Cisco Packet Tracer - D:\Year 2\Semester 1\Practicals\WEEK 3.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 1328, y: 174

2901 R-A 2901 R-B 1941 R-C

Time: 00:50:19

Search for device

Automatically Choose Connection Type

R-B

Physical Config CLI Attributes

IOS Command Line Interface

```
*LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
R-B(config)#exit
R-B
%SYS-5-CONFIG_I: Configured from console by console
R-B
R-B
R-B#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R-B(config)#router eigrp 100
R-B(config-router)#network 10.0.0.0
R-B(config-router)#network 192.168.1.0
R-B(config-router)#network 192.168.2.0
R-B(config-router)#network 192.168.3.0
R-B(config-router)#no auto-summary
R-B(config-router)#exit
%DUAL-5-NBRCCHANGE: IP-EIGRP 100: Neighbor 10.1.12.1 (GigabitEthernet0/0) is up: new adjacency
R-B(config)#
R-B(config)#end
R-B
%SYS-5-CONFIG_I: Configured from console by console
R-B
R-B#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H Address Interface Hold Uptime SRTT RTO Q Seq
(sec) (ms) Cnt Num
0 10.1.12.1 Gig0/0 13 00:05:04 40 1000 0 5
R-B
R-B#show ip route eigrp
10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
D 10.1.13.0/30 [90/3072] via 10.1.12.1, 00:05:11, GigabitEthernet0/0
D 192.168.1.0/24 [90/130816] via 10.1.12.1, 00:05:11, GigabitEthernet0/0
R-B
```

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20°C Partly sunny 1:27 PM 8/28/2026

Cisco Packet Tracer - D:\Year 2\Semester 1\Practicals\WEEK 3.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 1328, y: 174

2901 R-A 2901 R-B 1941 R-C

Time: 00:50:19

Search for device

Automatically Choose Connection Type

R-B

Physical Config CLI Attributes

IOS Command Line Interface

```
*LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
R-B(config)#exit
R-B
%SYS-5-CONFIG_I: Configured from console by console
R-B
R-B
R-B#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R-B(config)#router eigrp 100
R-B(config-router)#network 10.0.0.0
R-B(config-router)#network 192.168.1.0
R-B(config-router)#network 192.168.2.0
R-B(config-router)#network 192.168.3.0
R-B(config-router)#no auto-summary
R-B(config-router)#exit
%DUAL-5-NBRCCHANGE: IP-EIGRP 100: Neighbor 10.1.12.1 (GigabitEthernet0/0) is up: new adjacency
R-B(config)#
R-B(config)#end
R-B
%SYS-5-CONFIG_I: Configured from console by console
R-B
R-B#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H Address Interface Hold Uptime SRTT RTO Q Seq
(sec) (ms) Cnt Num
0 10.1.12.1 Gig0/0 13 00:05:04 40 1000 0 5
R-B
R-B#show ip route eigrp
10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
D 10.1.13.0/30 [90/3072] via 10.1.12.1, 00:05:11, GigabitEthernet0/0
D 192.168.1.0/24 [90/130816] via 10.1.12.1, 00:05:11, GigabitEthernet0/0
R-B
```

Copy Paste

20°C Partly sunny 1:27 PM 8/28/2026

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File Edit Options View Tools Extensions Window Help

Logical Physical x: 1348, y: 93

2901 R-A 2901 R-B 1941 R-C

Time: 00:50:51

Search for device

Automatically Choose Connection Type

R-C

Physical Config CLI Attributes

IOS Command Line Interface

```
R-C(config-if)# ip address 192.168.3.1 255.255.255.0
R-C(config-if)# exit
R-C(config)#
R-C(config)# end
R-C#
R-C#ping neighbor 192.168.3.1
R-C#
R-C#show ip eigrp neighbors
R-C#
R-C#show ip route eigrp
```

R-A

Physical Config CLI Attributes

IOS Command Line Interface

```
R-A(config-router)# network 192.168.3.0
R-A(config-router)# no auto-summary
R-A(config-router)# exit
R-A(config)#
R-A(config)# end
R-A#
R-A#show ip eigrp neighbors
R-A#
R-A#show ip route eigrp
```

pg. 7

Cisco Packet Tracer - D:\Year 2\Semester 1\Practicals\WEEK 3.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 1343, y: 81

Time: 00:52:20

Search for device

Automatically Choose Connection Type

2901 R-A

2901 R-B

1941 R-C

R-B

Physical Config CLI Attributes

IOS Command Line Interface

```
R-B(config-router)# network 192.168.3.0
R-B(config-router)# network 192.168.3.0
R-B(config-router)# no auto-summary
R-B(config-router)# exit
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.1.12.1 (GigabitEthernet0/0) is up: new adjacency
R-B(config)#
R-B(config)#end
R-B#
%SYS-5-CONFIG_I: Configured from console by console
R-B#
R-B#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H Address Interface Hold Uptime SRTT RTO Q Seq
0 10.1.12.1 Gig0/0 13 00:05:04 40 1000 0 5
R-B#
R-B#show ip route eigrp
10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
D 10.1.13.0/30 [90/3072] via 10.1.12.1, 00:05:11, GigabitEthernet0/0
D 192.168.1.0/24 [90/130816] via 10.1.12.1, 00:05:11, GigabitEthernet0/0
R-B#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.1.12.1 (GigabitEthernet0/0) is down: Interface Goodbye received
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.1.12.1 (GigabitEthernet0/0) is up: new adjacency
R-B#
R-B#
R-B#show ip route eigrp
10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
D 10.1.13.0/30 [90/3072] via 10.1.12.1, 00:00:44, GigabitEthernet0/0
D 192.168.0.0/16 [90/130816] via 10.1.12.1, 00:00:44, GigabitEthernet0/0
R-B#
```

R-A

Physical Config CLI Attributes

IOS Command Line Interface

```
0 10.1.12.2 Gig0/0 12 00:05:23 40 1000 0 5
R-A#
R-A#show ip route eigrp
10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
D 10.1.23.0/30 [90/3072] via 10.1.12.2, 00:03:50, GigabitEthernet0/0
D 192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
D 192.168.2.0/24 [90/130816] via 10.1.12.2, 00:03:50, GigabitEthernet0/0
R-A#
R-A#
R-A#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R-A(config)#interface g0/0
R-A(config-if)# ip summary-address eigrp 100 192.168.0.0 255.255.0.0
R-A(config-if)# exit
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.1.12.2 (GigabitEthernet0/0) is up: new adjacency
R-A(config)#
R-A(config)#end
R-A#
%SYS-5-CONFIG_I: Configured from console by console
R-A#
R-A#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R-A(config)#interface g0/0
R-A(config-if)# shutdown
R-A(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.1.12.2 (GigabitEthernet0/0) is down: interface down
R-A(config-if)#
```

Cisco Packet Tracer - D:\Year 2\Semester 1\Practicals\WEEK 3.plt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 1393, y: 140

Time: 00:55:13

Search for device

Automatically Choose Connection Type

R-B

Physical Config CLI Attributes

IOS Command Line Interface

```
R-B#
R-B#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H Address Interface Hold Uptime SRTT RTO Q Seq
0 10.1.12.1 Gig0/0 13 00:05:04 40 1000 0 5

R-B#
R-B#show ip route eigrp
10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
D 10.1.13.0/30 [90/3072] via 10.1.12.1, 00:05:11, GigabitEthernet0/0
D 192.168.1.0/24 [90/130816] via 10.1.12.1, 00:05:11, GigabitEthernet0/0

R-B#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.1.12.1 (GigabitEthernet0/0) is down:
Interface Goodbye received
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.1.12.1 (GigabitEthernet0/0) is up: new
adjacency

R-B#
R-B#
R-B#show ip route eigrp
10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
D 10.1.13.0/30 [90/3072] via 10.1.12.1, 00:00:44, GigabitEthernet0/0
D 192.168.0.0/16 [90/130816] via 10.1.12.1, 00:00:44, GigabitEthernet0/0

R-B#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down

R-B#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.1.12.1 (GigabitEthernet0/0) is down:
interface down

R-B#
R-B#show ip route eigrp
R-B#
```

R-A

Physical Config CLI Attributes

IOS Command Line Interface

```
R-A#
R-A#
R-A#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R-A(config)#interface g0/0
R-A(config-if)#ip summary-address eigrp 100 192.168.0.0 255.255.0.0
R-A(config-if)#exit
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.1.12.2 (GigabitEthernet0/0) is up: new
adjacency

R-A(config)#
R-A(config)#end
R-A#
$SYS-$-CONFIG_I: Configured from console by console

R-A#
R-A#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R-A(config)#interface g0/0
R-A(config-if)# shutdown

R-A(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.1.12.2 (GigabitEthernet0/0) is down:
interface down

R-A(config-if)#no shutdown

R-A(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.1.12.2 (GigabitEthernet0/0) is up: new
adjacency

R-A(config-if)#
```



Reflection Questions:

1. What's the practical difference between an OSPF inter-area route (O IA) and an intra-area route (O)?

O = Intra-area route. Learned inside the same area. OSPF trusts it more, it has full detail of the topology via LSAs Type 1 & 2.

O IA = Inter-area route. Learned from another area via ABR using LSA Type 3 summary. The ABR hides the details of Area 1 from Area 0. This scales the network - R1 doesn't need to know every link in Area 1, just that R-EDGE can get there. O is preferred over O IA if both exist.

2. In your test, did EIGRP or OSPF reconverge faster after a link failure? What does that tell you about distance-vector vs link-state routing?

In my test, EIGRP reconverged faster than OSPF single-area/multi-area.

EIGRP is Advanced Distance-Vector with DUAL algorithm - it keeps a Feasible Successor [backup route] in topology table already loop-free, so it can failover instantly without re-running SPF.

OSPF is Link-State - when a link fails, it must flood LSAs to all routers in the area and every router must re-run SPF calculation. This takes longer, but OSPF gives a complete map of the network and scales better with areas.